

Math 8 — End of Q4 Test

End-of-Q4 Content · 20 questions

Name: _____ Class: _____ Date: _____

Directions: Show all work. Box or circle your final answer. Include units where appropriate. Answers without units are accepted.

1. $v = \text{flambda}$. $f = 10 \text{ Hz}$, $\text{lambda} = 2 \text{ m}$. Speed? Answer: _____
2. $v = 340 \text{ m/s}$, $f = 170 \text{ Hz}$. lambda ? Answer: _____
3. $v = 3 \times 10^8 \text{ m/s}$, $\text{lambda} = 500 \text{ nm}$. Frequency? SN. Answer: _____
4. $\text{lambda} = 5 \text{ m}$, $v = 25 \text{ m/s}$. f ? Answer: _____
5. Light $3 \times 10^8 \text{ m/s}$. Time to travel $1.5 \times 10^{11} \text{ m}$? Answer: _____
6. Angle of incidence 30° . Reflection? Answer: _____
7. Convex lens: object beyond $2f$. Image is ____? Answer: _____
8. $T^2 = r^3$. If T doubles, r ratio? Answer: _____
9. $r = 4 \text{ AU}$. T in years? Answer: _____
10. $r = 9 \text{ AU}$. T in years? Answer: _____
11. Cylinder $r = 3$, $h = 5$. Volume ($\times \pi$). Answer: _____
12. Cone $r = 3$, $h = 4$. Volume. Answer: _____
13. Sphere $r = 3$. Volume. Answer: _____
14. Cylinder $r = 2$, $h = 10$. Answer: _____
15. Sphere $r = 6$. Answer: _____
16. Kai rides 12 mph for 4 hrs . Distance? Answer: _____
17. 15 m ladder, base 9 m . Height? Answer: _____
18. Pool $10 \times 5 \times 2 \text{ m}$. Volume? Answer: _____
19. 12 V , 3 A . P ? Answer: _____
20. $\$15/\text{hr} \times 40 \text{ hrs}$. Pay? Answer: _____